

Worksheet 9

October 5, 2026

Name: _____

About 15 minutes total. Begin independently, then discuss. Show your work.

1. Long division and an oblique asymptote

Let

$$R(x) = \frac{x^2 + 2x + 4}{x + 1}.$$

(a) Use polynomial long division to find the quotient and remainder.

(b) Rewrite R as quotient plus remainder over divisor. Give its oblique asymptote.

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2. Rational graph with a hole

Let

$$R(x) = \frac{x^2 - 4}{x^2 - x - 2}.$$

(a) Simplify and state all excluded x -values. Find the hole.

(b) Find the intercepts and all vertical/horizontal asymptotes.

(c) Sketch the graph, marking the hole with an open circle and labeling the asymptotes and intercepts.